ENHANCING HPC USE IN AFRICA THROUGH COLLABORATION WITHIN AFRICA:

A PILOT STUDY BETWEEN EAST AND WEST AFRICAN COUNTRIES

Co-designing People, Access, Workflows and Governance for a Stronger African HPC Ecosystem

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VIEW THE PROJECT
(GitHub Repository)

https://github.com/karegapauline/BHKi-projects/tree/main/HPC_project

1. INTRODUCTION

- High-performance computing (HPC) is increasingly central to modern bioinformatics, genomics, public health, and environmental research.
- Despite the presence of multiple HPC systems across Africa, evidence suggests persistent underutilization, restricted access, and misalignment between available computational resources and researchers' needs.
- Additionally, this has been reported multiple times through multiple studies, small and large scale. However, solutions remain unpublished or unimplemented.
- We reviewed the existing literature to identify key findings and co-design a framework that strengthens collaboration between regions with complementary strengths and needs.

2. WHY KENYA – WEST AFRICA?

Increase Joint Capacity

- Complementary strengths
- Resource sharing
- Cross-learning

Strong Individual Communities

- Active HPC initiatives
- Domain expertise
- Growing user bases

WEST AFRICA

- PUBREC (Benin)
- WACREN
- PATHER
- Multiple university clusters

KENYA

- H3ABioNet
- KENET
- icipe HPC
- University clusters

3. KEY CHALLENGES IDENTIFIED ACROSS AFRICAN HPC ECOSYSTEMS

CAPACITY

- Institution-based, small clusters with limited storage and compute
- Few national investments in HPC
- Reliance on grants and cloud platforms

SKILLS

- Lack of HPC education and basic understanding
- Need for project- and problem-based training
- Shortage of technical personnel
- Educational background and history gaps

ACCESS

- Access restricted to institutional or funded projects
- Limited training and use for external researchers/students
- Low awareness leads to underutilization

COMMUNITY

- Fragmented, domain-specific communities
- Poor visibility of resources
- Weak connection between communities across regions
- Need for leadership support and recognition

> Better HPC in Africa is not only about building larger clusters—
> it is about co-designing access, training, governance, and community around existing infrastructure.

4. THE PROBLEM WE AIM TO SOLVE

RESEARCHER / RESOURCE NEEDER

- Lack of knowledge resource exists
- Lack of skills
- Access issues
- Limited knowledge
- Poor communication of their needs

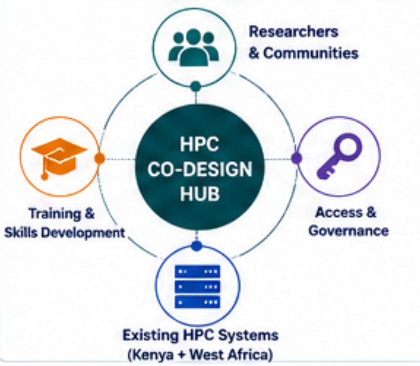
HPC RESOURCE HAVER

- Underutilization of resources
- Lack of well trained personnel
- Poor maintenance

TOOLS TO FACILITATE BETTER COMMUNICATION:
Compute Resource Management Plan

TOOLS TO UNDERSTAND EACH OTHER BETTER:
Learning Material & Skills Development

5. OUR PROPOSAL: HPC CO-DESIGN FRAMEWORK



- A shift from infrastructure-centric thinking to HPC co-design around research workflows, skills, governance and sustainability.
- This pilot links Kenya and selected West African institutions by reconfiguring access, workflows and governance across existing systems.
- Training, skills development, workflow optimization, and access/governance are treated as integral components of HPC infrastructure.
- BHKi acts as a community-led steering entity connecting researchers, infrastructure providers and funders.

6. PHASE 1: CO-DESIGN ACTIVITIES

1. Co-design training materials for pre-use knowledge (building basics, resource needs, running and submitting jobs, loading modules).
2. Co-design access protocols to accommodate more students without institutional access.
3. Test new ideas such as DMPs or OSF pre-registration to request access and as a tool to teach resource assessment.
4. Co-design hackathons and competitions to build and sharpen HPC skills among practicing communities.
5. Co-design a support system that assists users in understanding their needs, regardless of affiliation and region (sustained via train-the-trainer programs).

7. SUGGESTED APPROACHES FROM LITERATURE

- **Hybridization model** (not published): mapping resources and recommending continental & regional coordination, detailed inventories and access models.
- **UKRI–UVRI grant:** funding access to UVRI HPC and training.
- Many solutions exist but remain unimplemented, as they are often designed for organizations, not communities.

8. THE HOW – IMPLEMENTATION ROADMAP

1. **Workshop of HPC system admins from multiple universities and institutions**
   Discussion on how to better support users and become trainers (train-the-trainer).
2. **2–3 Sprints**
   Gather, analyze and collate commonly used training materials to identify gaps and create user-ready content.
3. **Major Stakeholder Workshops**
   Co-design access protocols, governance mechanisms and collaborative workflows.
4. **Surveys & Focus Groups for HPC Users**
   Assess skills, training gaps, access barriers and test new materials with feedback for continuous improvement.

9. EXPECTED IMPACT

From Underutilized HPC Resources → To Highly Utilized Resources → From Limited & Restricted Access → To Regional Access Models → From Fragmented Communities → To Connected African HPC Network

Stronger Skills & Workforce | Better Research Outcomes | Collaborative Innovation | Sustainable HPC Ecosystem in Africa

"By co-designing together, we can build an inclusive, sustainable and collaborative HPC future for Africa."